Lab 5 Report Microprocessor

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1 LCD Interfacing with Pt-51 Microcontroller

1.1 Objective

To understand the electrical and logical connection between a standard 16 × 2 Character LCD and the 8051-based Pt-51 microcontroller kit.To learn how to integrate and compile external header files (lcd.h) and source files (lcd.c) within an embedded C environment. To modify existing code to display custom alphanumeric strings (”EE-337” and a personal name) while implementing basic formatting like centering via space padding.

1.2 Observation

- Initial State: Upon loading the default hex file, the LCD successfully initializes and displays the hardcoded strings:
  Line 1: Pt-51
  Line 2: IIT Bombay
- Functionality: lcd.h functions like lcd_init(), lcd_cmd(), and lcd_write_string()) correctly handle the timing requirements and "Enable" pulses required by the LCD controller
- Modified Output: After modifying lcd.c, the LCD clears the previous content and displays the new strings centered on their respective lines.
  Line 1: EE337
  Line 2: UMESH

1.3 Conclusion

Successful Interfacing: The experiment confirms that the Pt-51 kit can effectively communicate with peripheral display modules through a defined set of GPIO pins.
Abstraction via Headers: By using lcd.h, the complexity of low-level bit manipulation (setting RS, RW, and EN pins) is abstracted into high-level C functions, making the code more readable and easier to maintain.

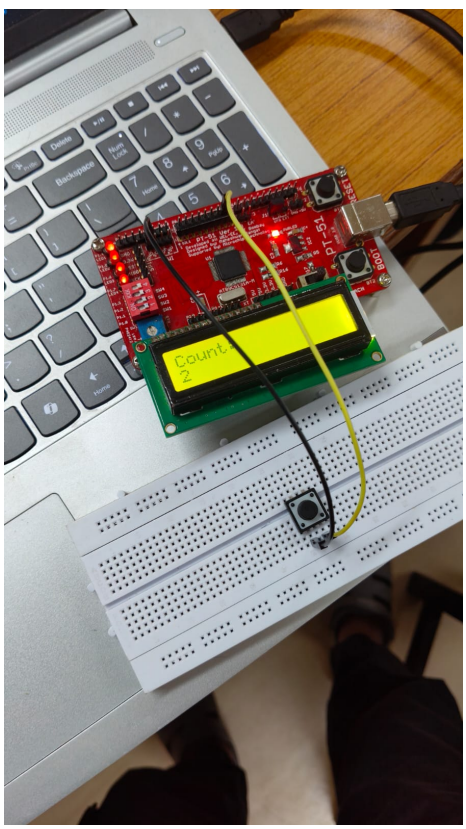
2 Key Debouncing and Pulse Counting on Pt-51

2.1 Objective

To design and implement an embedded C program for the 8051-based Pt-51 kit that:

- Configures P0.7 as an input pin for a push-button
- Implements a software debouncing algorithm to filter out mechanical switch noise and Accurately increments a counter variable only upon a valid, stable button press
- Displays the real-time counter value on a 16×2 LCD using custom character conversion logic

2.2 Results



2.3 Observations

- Upon system reset, the LCD initializes and displays the static string "Button Count:" on the first line and "0" on the second line
- During the button press, the program waits for a stable period (20 ms). If the button is released faster than this duration (simulating noise or a "flick"), the counter does not increment, confirming the debounce logic is functional
- Because the code includes the while(button == 0); loop, holding the button down for several seconds results in only one increment. The counter only advances again after the button is released and pressed a second time
- Since the LCD controller accepts ASCII values, the software successfully performs integer-to-character conversion using the modulus and division operators (e.g., count
- When the count increases from 9 to 10 or 99 to 100, the display correctly updates the second line without leaving "ghost" digits from previous values, thanks to the space padding (" ") used in the code

2.4 Conclusion

- The experiment proves that a software-controlled delay is a highly effective method for eliminating "double-counting" caused by the physical properties of mechanical switches.
- By utilizing the at89c5131.h and lcd.h libraries, complex hardware interactions are simplified into logical C structures
- Proper LCD management requires not just sending data, but also managing the cursor position and clearing/padding lines to ensure the user interface remains readable and centered